Optimizing Homomorphic Encryption in Federated Learning with Zero-Skipping

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***Abstract*—Federated learning enables training AI model without sharing of data, but requires protections such as homomorphic encryption to prevent reverse engineering of shared gradients, which incurs significant computational overhead. To mitigate this overhead, we propose an improved compression scheme for homomorphic encryption that integrates two key optimizations: clipped quantization and zero-skipping encryption. Clipped quantization increases the proportion of zeros during gradient quantization, while zero-skipping encryption identifies blocks that are quantized to zero and omits homomorphic encryption, thereby reducing the overhead of homomorphic encryption. The preliminary experiments for the federated learning with CNN show that the improved homomorphic encryption method could reduce client-side homomorphic encryption time and overall training time with a small decrease in accuracy.**

***Index Terms*—federated learning, homomorphic encryption, clipped quantization, zero skipping.**

I. Introduction

Federated learning (FL) enables multiple clients to collaboratively train a shared model by exchanging model updates or gradients without sharing raw data, making it suitable for privacy-sensitive domains such as healthcare, finance, and smart cities under regulations like GDPR and HIPAA [1]. However, even without data sharing, unencrypted gradient exchange remains vulnerable to privacy attacks such as model inversion and membership inference [2]. To mitigate these risks, prior work has incorporated homomorphic encryption (HE) into FL, allowing secure aggregation of encrypted gradients on the server [3]. Despite its strong privacy guarantees, HE introduces substantial computational overhead, significantly limiting its practical deployment.

To reduce this overhead, several studies have proposed optimization techniques for HE-based FL [4]–[6]. Among them, BatchCrypt [4] is a state-of-the-art approach that quantizes gradients and packs multiple values into a single ciphertext, substantially reducing the number of HE operations. However, BatchCrypt still encrypts all quantized gradient components, including those that are zero or negligibly small. Since deep neural networks typically produce sparse gradients—especially after quantization—encrypting blocks consisting entirely of zeros leads to unnecessary computation and increased communication cost, thereby hindering scalability.

To address this inefficiency, we propose an enhanced BatchCrypt framework that improves the skip ratio, defined as the proportion of quantized blocks that can be omitted from encryption due to containing only zero or negligible values. By selectively skipping such blocks, the proposed method reduces client-side encryption overhead and ciphertext transmission volume without degrading aggregation correctness or model accuracy. Furthermore, we explore clipped quantization techniques to increase gradient sparsity, thereby further improving the effectiveness of the skipping mechanism.

II. Zero-Skipping Homomorphic Encryption in FL

Homomorphic encryption (HE) is a powerful cryptographic fundamental that allows computations to be performed directly on encrypted data without prior decryption. This property is particularly important in privacy-preserving machine learning, as it enables sensitive data to remain encrypted throughout the entire processing phase, including intermediate computations. Fully homomorphic encryption (FHE) supports arbitrary computations on encrypted data, but its high algorithmic complexity and substantial computational overhead generally limit its practical application in large-scale systems. Therefore, in scenarios where computations primarily involve aggregation operations, such as gradient aggregation in federated learning (FL), additive homomorphic encryption (AHE) offers a more practical and efficient solution with lower overhead.

BatchCrypt is a privacy-preserving FL framework specifically designed to securely and efficiently aggregate local gradient updates from multiple clients using AHE. In BatchCrypt, each client independently performs local model training on its private dataset and subsequently computes local gradient updates. Prior to uploading their gradient updates, clients encrypt these gradients using the server's public key to prevent information leakage during transmission and aggregation. Due to the computational overhead associated with AHE schemes, gradients are typically quantized and packed efficiently into ciphertexts. BatchCrypt employs this quantization and batching strategy to reduce the number of ciphertexts and minimize computational and communication costs.

While the original BatchCrypt framework provides a solid foundation for privacy-preserving FL, our analysis identifies

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) [RS-2025-25463940]

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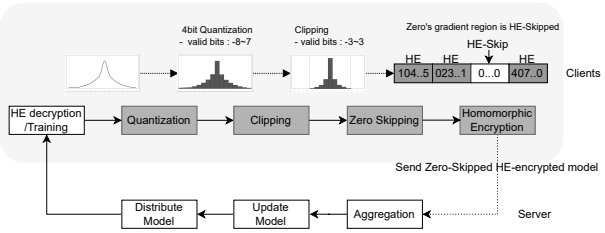

Fig. 1. The architecture and operation of the federated learning system that applies HE based on Zero-Skipping

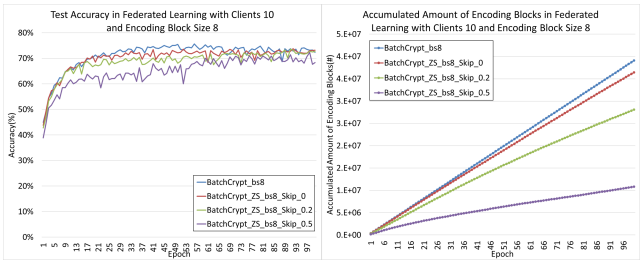

Fig. 2. Experimental Results for Alexnet in Federated Learning system with Homomorphic Encoding; 10 Clients and HE Block size with 8.

clear opportunities for further optimization, especially in terms of client-side encryption workload and overall aggregation latency. In this paper, we identify inefficiencies in existing HE-based FL frameworks, particularly those related to the encryption of redundant zero or near-zero gradient data, and propose an improved BatchCrypt framework, called BatchCrypt-ZS. Our goal is to significantly reduce encryption overhead in privacy-preserving FL while maintaining model accuracy and privacy guarantees comparable to the original framework.

To achieve this goal, we introduce optimization strategies with zero-skipping homomorphic encryption for clipped quantization. The architecture and operation of the FL system applying HE based on zero-skipping are shown in Figure reffig:arch. The FL system first performs quantization on locally trained gradients using the standard BatchCrypt quantization scheme. Next, a clipping operation is applied to further increase the proportion of zero-valued elements in the quantized gradients. This process enhances gradient sparsity and increases the zero ratio across gradient blocks. During the HE stage, encryption is performed only for non-zero-dominated blocks. Since a large fraction of gradients quantize precisely to zero, encrypting zero-only blocks is computationally redundant. Therefore, BatchCrypt-ZS skips the encryption of such blocks, directly reducing the client-side encryption workload and improving efficiency in HE-based federated learning.

III. Experiments

We present a preliminary evaluation of the proposed enhanced BatchCrypt framework, termed BatchCrypt-ZS, focusing on accuracy, communication overhead, and computational efficiency. BatchCrypt-ZS was implemented within a BatchCrypt-based FL system using Paillier homomorphic encryption [3] and compared against the baseline BatchCrypt approach. Experiments were conducted using the AlexNet [7] model on the CIFAR-10 dataset with the RMSprop optimizer. The FL system consisted of 10 clients and one server, trained for 100 epochs. Gradients were quantized to 8-bit integers, and homomorphic encryption was applied by packing eight quantized values into a single batch block. We evaluated BatchCrypt-ZS with skip-threshold values of 0, 0.2, and 0.5, corresponding to cases where 0%, 20%, and 50% of gradient values within a block are zero, respectively.

Figure 2 presents the experimental results, where the left plot shows test accuracy and the right plot shows HE efficiency. As the skip-threshold increases, BatchCrypt-ZS shows slightly slower accuracy improvement in early epochs due to a higher proportion of skipped gradient blocks. However, this accuracy gap gradually diminishes and becomes negligible as training progresses, indicating that skipping zero-dominated blocks has minimal impact on final model accuracy. In contrast, BatchCrypt-ZS significantly improves system efficiency by reducing encryption workload and communication overhead. As shown in the right plot, the accumulated number of HE-encoded gradient blocks decreases with higher skip-thresholds, leading to lower encoding time and reduced transmission volume, thereby enhancing overall FL system efficiency.

IV. Conclusion

In FL, sharing unencrypted gradients can leak sensitive information, motivating the need for efficient homomorphic encryption–based methods that preserve privacy without excessive computational overhead. BatchCrypt-ZS is an enhanced HE-based FL framework that improves computational and communication efficiency through zero-skipping and clipping-based quantization. Experimental results show that BatchCrypt-ZS effectively balances privacy preservation and efficiency for FL systems.

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